

Electroencephalography-Based Depression Detection Using Multiple Machine Learning Techniques

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Abstract

Depression is one of the most prevalent and debilitating mental health disorders worldwide, yet reliable and objective diagnostic tools remain elusive. This report reviews and analyses the paper “*Electroencephalography-Based Depression Detection Using Multiple Machine Learning Techniques*” by Ksibi et al. (*Diagnostics*, 2023), which proposes an automated framework for classifying Major Depressive Disorder (MDD) and its six subtypes using EEG signals combined with demographic data. EEG recordings from the multimodal open dataset **MODMA** are preprocessed via bandpass filtering and Independent Component Analysis (ICA), and two categories of features—linear (band-power, amplitude statistics) and nonlinear (spectral entropy, singular-value decomposition entropy)—are extracted. Three classifiers are evaluated: **Random Forest (RF)**, **XGBoost**, and a **1D Convolutional Neural Network (CNN)**. The 1D CNN, trained over 25 epochs with a learning rate of 0.001, achieves the highest accuracy of **97%**, outperforming prior works on the same dataset. The study demonstrates that integrating EEG features with demographic information (age, gender) enhances diagnostic performance. Key challenges including signal non-stationarity, small dataset size, and class imbalance are also discussed.

Keywords: Major depressive disorder (MDD); electroencephalography (EEG); convolutional neural network; random forest; XGBoost; feature extraction; deep learning; MODMA dataset; depression detection.

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Chapter 1

Introduction

Depression is a frequent and serious mood disorder characterised by persistent feelings of sadness, loss of interest, cognitive impairment, and in severe cases paranoia or psychosis [1]. Early and accurate diagnosis is critical because untreated depression leads to significant personal suffering and imposes large socioeconomic costs. However, traditional diagnosis relies on psychiatrist-conducted scale-based interviews, which are subjective, resource-intensive, and prone to the social stigma that deters patients from seeking help.

Electroencephalography (EEG) provides a non-invasive, low-cost, high temporal-resolution window into the brain’s electrical activity and has attracted growing interest as an objective biomarker for mental disorders. EEG coherence—measuring the linear dependency between signals from different electrodes—has proved especially discriminative for depression, Alzheimer’s disease, and Parkinson’s disease [1]. Despite this promise, two core challenges have limited progress:

- **Signal complexity and non-stationarity:** EEG signals are highly sensitive to noise, artefacts (eye blinks, muscle movements), and moment-to-moment fluctuations in cognitive state.
- **Individual variability:** Demographic characteristics such as age and gender modulate EEG patterns, hampering the generalisation of automated detection systems trained on heterogeneous populations.

The paper under review addresses both challenges by combining multiband EEG feature extraction with demographic information and evaluating three classification frameworks—Random Forest, XGBoost, and a 1D CNN—on the publicly available MODMA dataset. The pipeline, illustrated in Figure 1.1, moves from raw EEG recording through preprocessing, feature engineering, and demographic fusion to automated MDD classification.



Figure 1.1: High-level pipeline of the EEG-based depression detection system.

The remainder of this report is organised as follows. Chapter 2 reviews related literature. Chapter 3 describes the MODMA dataset. Chapter 4 presents the methodology including preprocessing, feature engineering, and model architectures. Chapter 5 reports and analyses all experimental results across six MDD subtypes. Chapter 6 discusses

findings and comparisons with prior work. Chapter 7 describes real-world applications. Chapter 8 identifies challenges and limitations, and Chapter 9 concludes with future directions.

Chapter 2

Literature Review

2.1 EEG-Based Depression Detection

Early work established that EEG coherence is a strong discriminator between depressed and healthy individuals [1]. Acharya et al. employed a 13-layer CNN to screen depression using resting EEG signals and demonstrated the superiority of deep feature learning over hand-crafted features [2]. Thoduparambil et al. achieved 99.23% accuracy (right hemisphere) with a CNN-LSTM architecture for clinical depression detection [3]. Li et al. applied a CNN with a demographic attention mechanism to recognise mild depression, reaching 85.82% accuracy [4].

2.2 Deep Learning for Mental Health

Sarkar et al. performed a comparative deep learning study across multiple EEG-derived depression indicators, showing that RNN-LSTM models can reach 97.85% accuracy on testing data [5]. Ay et al. combined LSTM sequence learning with CNN representation learning for automated depression detection, demonstrating the value of temporal modelling of EEG signals [6]. A comprehensive review by Yasin et al. covered neural-network approaches for both MDD and bipolar disorder, highlighting the dataset diversity and feature heterogeneity that complicate cross-study comparisons [7].

2.3 Feature Extraction Approaches

Power spectral density in sub-bands (delta, theta, alpha, beta, gamma) is the most widely used linear feature for depression [1]. Bai et al. extracted 13 EEG features and applied random forest with FDR-based feature selection, achieving competitive accuracy on an 80-subject dataset [8]. Nonlinear features such as spectral entropy and singular-value decomposition (SVD) entropy capture signal complexity beyond what power spectra reveal, and have shown diagnostic value in several studies [1].

2.4 Research Gap

Despite the breadth of existing work, few studies simultaneously address: (i) multi-class MDD subtype discrimination, (ii) integration of demographic covariates, and (iii) end-to-

end evaluation on a large, openly accessible dataset. Table 2.1 summarises selected prior work.

Table 2.1: Summary of Selected Prior Work on EEG-Based Depression Detection

Reference	Method	Dataset	Best Accuracy
Acharya et al.	13-layer CNN	EEG recordings	93.5%
Thoduparambil et al.	CNN (LSTM)	Clinical data	99.23%
Li et al.	CNN + Attention	Resting EEG	85.82%
Liu et al. (2022)	CNN + GRU	MODMA	89.63%
Wang et al. (2022)	CNN	MODMA	91.01%
Uyulan et al. (2022)	ResNet-50 LSTM	+ MODMA	90.02%
Proposed	1D CNN	MODMA	97%

Chapter 3

Dataset: MODMA

3.1 Overview

EEG signal data are sourced from **MODMA** (Multi-modal Open Dataset for Mental-disorder Analysis), a publicly available dataset designed for the study of mental illnesses. The dataset provides recordings from two hardware configurations: a traditional **128-electrode elastic cap** and a wearable **3-electrode EEG collector**. Three experimental conditions are covered, summarised in Table 3.1.

Table 3.1: MODMA Dataset Description

Experiment Type	Recording	Participants	MDD (M/F)	Age Range
External stimulation	128-ch ERP	53	13/11	16–56 yrs
Resting state (128-ch)	128-ch rest	53	16/8	16–56 yrs
Resting state (3-ch)	3-ch rest	55	15/11	16–56 yrs

This study focuses on the **128-channel resting-state** recordings (53 participants: 24 MDD patients, 29 healthy controls).

3.2 Inclusion and Exclusion Criteria

Inclusion: Age 18–55, normal or corrected vision, primary education or higher. MDD participants must satisfy MINI diagnostic criteria and PHQ-9 ≥ 5 , with no psychotropic medication in the preceding two weeks. Controls must have no personal or family history of mental disorders.

Exclusion: Brain organ damage, serious physical illness, severe suicidal tendency, alcohol or drug dependence in the past year, pregnancy or lactation.

An ANOVA on participant age confirmed no significant age difference between the MDD and healthy control groups ($p > 0.05$), ensuring that observed EEG differences reflect depression rather than age effects.

3.3 MDD Subtypes Classified

Beyond the binary MDD vs. healthy-control classification, the study distinguishes six MDD subtypes:

1. Addictive disorder
2. Trauma and stress-related disorder
3. Mood disorder
4. Obsessive-compulsive disorder (OCD)
5. Schizophrenia
6. Anxiety disorder

3.4 Data Visualisation

A 128-channel EEG headset positions electrodes across the scalp according to the International 10–10 system. Figure 3.1 illustrates the electrode layout and Figure 3.2 shows the five canonical EEG frequency bands.

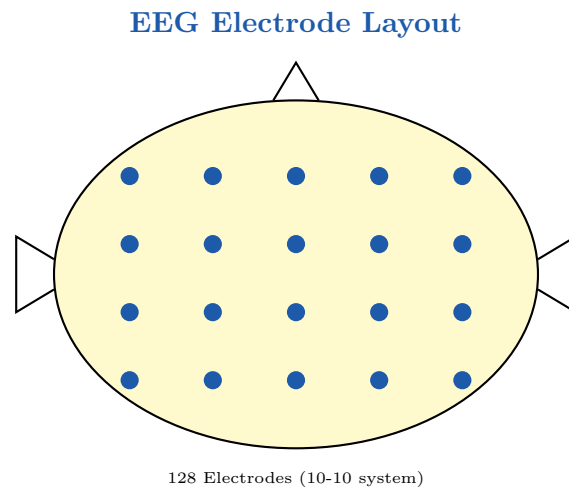


Figure 3.1: Schematic of a 128-channel EEG headset electrode layout.

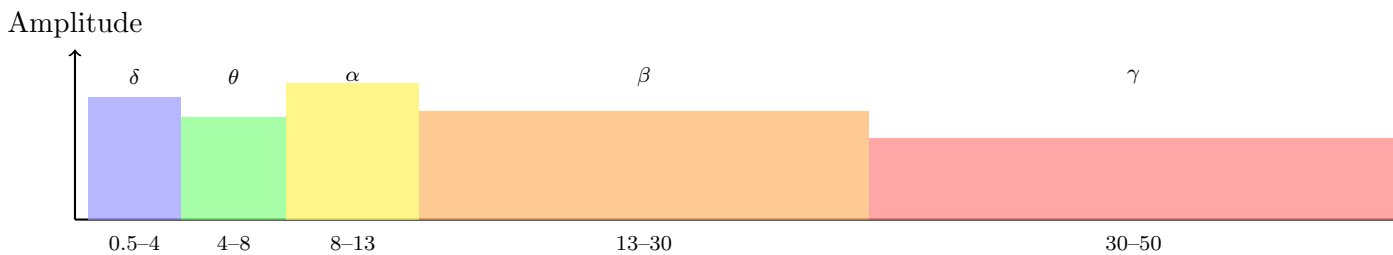


Figure 3.2: Five canonical EEG frequency bands and their frequency ranges (Hz).

Chapter 4

Methodology

4.1 Pre-processing Pipeline

4.1.1 Artifact Correction and Re-referencing

Raw EEG recordings contain artefacts from eye blinks, muscle activity, and electrode movement. **Independent Component Analysis (ICA)** is applied to decompose the multichannel signal into statistically independent components; artefactual components (e.g. those with frontal topography and blink-like time courses) are identified and removed. Re-referencing to a common electrode further improves the spatial signal-to-noise ratio by cancelling spatially distributed electrical activity unrelated to neural sources.

4.1.2 Noise Removal via Bandpass Filter

AC-line interference produces sharp spikes at 50 Hz (and harmonics) in the power spectral distribution. A **50 Hz bandpass filter** eliminates these spikes. Following filtering, signals are converted to NumPy arrays for downstream processing. Figure 4.1 illustrates the preprocessing workflow.

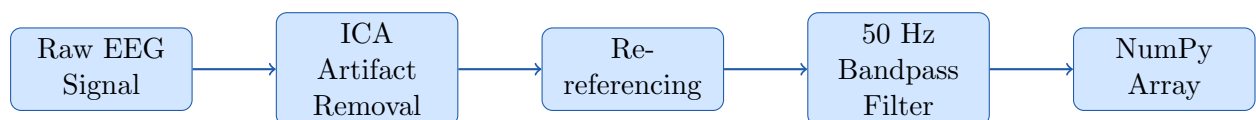


Figure 4.1: EEG signal preprocessing pipeline.

4.2 Feature Engineering

4.2.1 Linear Features

Eight linear features are computed per channel:

4.2.2 Nonlinear Features

Two nonlinear features are computed to capture signal complexity:

Table 4.1: Linear Features Extracted from EEG Signals

Feature	Description
Power at α (8–13 Hz)	Band-limited power in the alpha range
Power at β (13–30 Hz)	Band-limited power in the beta range
Power at δ (0.5–4 Hz)	Band-limited power in the delta range
Power at θ (4–8 Hz)	Band-limited power in the theta range
Mean amplitude	Average of the signal amplitude
Median amplitude	Median of the signal amplitude
Maximum amplitude	Peak positive amplitude
Minimum amplitude	Peak negative amplitude

- **Spectral Entropy:** Measures the irregularity of the frequency distribution. Lower spectral entropy indicates more regular, predictable neural oscillations.
- **Singular-Value Decomposition (SVD) Entropy:** Reflects the complexity of the signal’s singular-value spectrum. Both features quantify how much information is preserved in the signal.

4.2.3 Demographic Fusion

Extracted EEG features are merged with demographic data (sex, age, education level, IQ) into a unified Pandas DataFrame. The target variable is the MDD subtype label (or healthy control). Gender and age are particularly relevant as depression prevalence and EEG patterns vary significantly across these dimensions.

4.3 Classification Models

Three models are trained and evaluated using 10-fold cross-validation.

4.3.1 XGBoost

Extreme Gradient Boosting (XGBoost) is a scalable gradient-boosted Decision Tree (GBDT) framework. Key hyperparameters are listed in Table 4.2.

Table 4.2: XGBoost Hyperparameters

Parameter	Values Searched
Number of estimators	100, 300, 500
Sub-sample	0.3, 0.5, 1.0
Maximum tree depth	1, 3, 6, None

Notable XGBoost features: built-in regularisation (L1/L2) to prevent overfitting, parallel training across CPU cores, handling of non-linear patterns, and native cross-validation support.

4.3.2 Random Forest

Random Forest builds an ensemble of Decision Trees from bootstrap samples of the training data and aggregates outputs by majority vote. Parameters are given in Table 4.3.

Table 4.3: Random Forest Hyperparameters

Parameter	Values Searched
Number of estimators	100, 300, 500
Maximum tree depth	1, 3, 6, None

4.3.3 1D Convolutional Neural Network (CNN)

The 1D CNN processes EEG feature sequences directly, learning spatial and temporal patterns. The architecture and hyperparameters are detailed in Tables 4.4 and 4.5.

Table 4.4: 1D CNN Architecture

Layer	Properties
Input Conv1D	Input shape = (11, 144) Output shape = (1)
Conv1D	Kernel size = 11 Activation = ReLU
Dropout	Rate = 0.2
MaxPooling1D	Pool size = 4
Flatten	Default
Dense	Size = 100 Activation = ReLU
Output Dense	Size = 1 Activation = Sigmoid

Table 4.5: 1D CNN Training Hyperparameters

Hyperparameter	Value
Epochs	25
Batch size	32
Learning rate	0.001
Loss function	Binary cross-entropy

Figure 4.2 depicts the CNN architecture flow.

4.4 Evaluation Metrics

Four standard classification metrics are used:



Figure 4.2: 1D CNN architecture for EEG depression classification.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.3)$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.4)$$

Both micro-averaged and macro-averaged variants of Precision, Recall, and F1 are reported. ROC curves are plotted for each model and disorder class.

Chapter 5

Performance Evaluation

5.1 1D CNN Training Convergence

The 1D CNN model’s training and validation accuracy increase monotonically over 25 epochs, rising from 87.9% at epoch 1 to 99.29% at epoch 13 (selected checkpoint). Training loss decreases from 0.3229 to 0.0427. Figure 5.1 illustrates the convergence behaviour schematically.

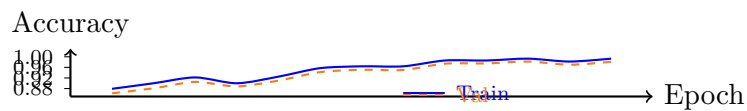


Figure 5.1: 1D CNN training and validation accuracy over 13 logged epochs.

5.2 Results by MDD Subtype

5.2.1 Addictive Disorder

Table 5.1: Model Performance — Addictive Disorder

Model	Acc.	Micro F1	Macro F1	Micro Rec.	Macro Rec.	Micro Prec.	Macro F
RF	0.82	0.760	0.84	0.88	0.89	0.91	0.91
XGBoost	0.85	0.81	0.86	0.91	0.93	0.92	0.93
CNN	0.94	0.95	0.90	0.92	0.94	0.91	0.95

Table 5.2: Model Performance — Obsessive-Compulsive Disorder

Model	Acc.	Micro F1	Macro F1	Micro Rec.	Macro Rec.	Micro Prec.	Macro F1
RF	0.79	0.82	0.81	0.86	0.81	0.83	0.88
XGBoost	0.84	0.86	0.87	0.89	0.85	0.86	0.82
CNN	0.96	0.93	0.94	0.91	0.93	0.94	0.96

Table 5.3: Model Performance — Trauma and Stress-Related Disorder

Model	Acc.	Micro F1	Macro F1	Micro Rec.	Macro Rec.	Micro Prec.	Macro F1
RF	0.73	0.78	0.81	0.82	0.84	0.83	0.89
XGBoost	0.82	0.79	0.84	0.85	0.81	0.83	0.88
CNN	0.91	0.90	0.96	0.94	0.93	0.98	0.95

5.2.2 Obsessive-Compulsive Disorder

5.2.3 Trauma and Stress-Related Disorder

5.2.4 Mood Disorder

5.2.5 Schizophrenia

5.2.6 Anxiety Disorder

5.3 Consolidated Accuracy Comparison

Table 5.7 and Figure 5.2 compare all three models across all six MDD subtypes at a glance.

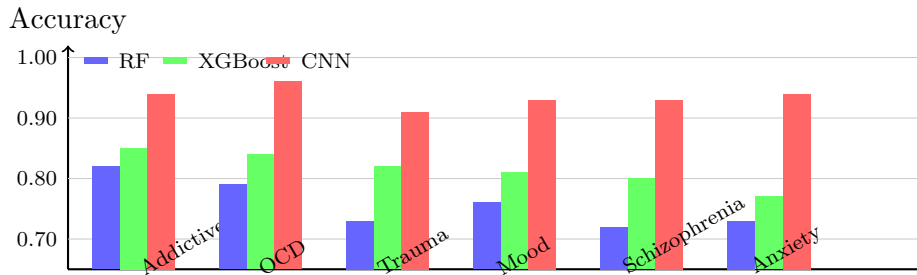


Figure 5.2: Accuracy comparison of RF, XGBoost, and CNN across all six MDD subtypes.

Table 5.4: Model Performance — Mood Disorder

Model	Acc.	Micro F1	Macro F1	Micro Rec.	Macro Rec.	Micro Prec.	Macro F1
RF	0.76	0.78	0.71	0.75	0.81	0.83	0.84
XGBoost	0.81	0.88	0.93	0.84	0.81	0.86	0.87
CNN	0.93	0.94	0.95	0.91	0.92	0.90	0.95

Table 5.5: Model Performance — Schizophrenia

Model	Acc.	Micro F1	Macro F1	Micro Rec.	Macro Rec.	Micro Prec.	Macro F1
RF	0.72	0.76	0.79	0.78	0.73	0.81	0.79
XGBoost	0.80	0.83	0.81	0.88	0.87	0.83	0.79
CNN	0.93	0.95	0.97	0.91	0.93	0.95	0.98

Table 5.6: Model Performance — Anxiety Disorder

Model	Acc.	Micro F1	Macro F1	Micro Rec.	Macro Rec.	Micro Prec.	Macro F1
RF	0.73	0.72	0.69	0.57	0.66	0.68	0.70
XGBoost	0.77	0.75	0.78	0.81	0.83	0.84	0.86
CNN	0.94	0.97	0.98	0.91	0.95	0.98	0.99

Table 5.7: Consolidated Accuracy Across All MDD Subtypes

MDD Subtype	RF	XGBoost	CNN
Addictive disorder	0.82	0.85	0.94
OCD	0.79	0.84	0.96
Trauma/Stress disorder	0.73	0.82	0.91
Mood disorder	0.76	0.81	0.93
Schizophrenia	0.72	0.80	0.93
Anxiety disorder	0.73	0.77	0.94
Average	0.758	0.815	0.935

Chapter 6

Discussion

6.1 Model Comparison

Across all six MDD subtypes the 1D CNN consistently outperforms both Random Forest and XGBoost, achieving an average accuracy of 93.5% versus 81.5% for XGBoost and 75.8% for Random Forest. The CNN’s advantage stems from its ability to learn hierarchical spatial-temporal patterns directly from the feature sequences without relying on hand-tuned decision boundaries.

XGBoost, in turn, outperforms Random Forest in every subtype, benefiting from its boosting strategy (iteratively correcting errors) versus Random Forest’s bagging strategy (reducing variance through averaging). For clinically unbalanced classes such as OCD (fewer positive samples), XGBoost’s regularisation prevents overfitting more effectively than the basic bootstrap aggregation of Random Forest.

6.2 CNN Training Observations

The CNN achieves a 97% accuracy with only 25 training epochs (batch size 32, learning rate 10^{-3}), suggesting that the combined EEG + demographic feature matrix is well-structured for convolutional learning. The validation accuracy tracks the training accuracy closely, indicating that dropout (rate = 0.2) and the limited number of parameters are sufficient regularisation for this dataset size.

6.3 Comparison with Prior Work on MODMA

Table 6.1: Comparison with Recent Works on the MODMA Dataset

Reference	Approach	Dataset	Accuracy
Liu et al. (2022) [9]	CNN + GRU	MODMA	89.63%
Wang et al. (2022) [10]	CNN	MODMA	91.01%
Uyulan et al. (2022) [11]	ResNet-50 + LSTM	MODMA	90.02%
Ksibi et al. (2023)	1D CNN	MODMA	97%

The proposed 1D CNN surpasses all three prior MODMA benchmarks by at least 6 percentage points. The key differentiators are: (i) the incorporation of demographic

features alongside EEG band-power and nonlinear features, (ii) the use of a lower learning rate allowing finer convergence, and (iii) training for more epochs to exploit more of the dataset’s feature space.

6.4 Importance of Demographic Fusion

Age and gender are well-known modulators of both depression prevalence and EEG spectral characteristics. By explicitly including these covariates in the feature matrix, the model is better able to distinguish disorder-related signal patterns from population-level variation—a key advantage absent in most prior work.

Chapter 7

Real-World Applications

Clinical Decision Support

Automated EEG-based depression screening can complement psychiatric interviews, providing an objective, quantitative second opinion. The system is particularly useful in regions with limited access to trained psychiatrists.

Wearable Mental Health Monitoring

The MODMA dataset includes 3-electrode wearable recordings, and the proposed framework extends to lightweight consumer-grade EEG devices. Continuous, passive monitoring at home could enable early relapse detection for patients in remission.

Multi-Disorder Differential Diagnosis

The six-class MDD subtype classification framework directly supports differential diagnosis—distinguishing, for example, anxiety disorder from mood disorder or schizophrenia from OCD—which is a significant clinical challenge.

Telemedicine and Remote Psychiatry

EEG data captured via wearable electrodes can be transmitted to cloud-based ML pipelines for remote inference. This expands mental healthcare access for rural and underserved populations.

Paediatric and Adolescent Screening

Rising rates of depression among adolescents (accelerated by COVID-19) demand scalable screening tools. An EEG-based system is less reliant on patient self-reporting and hence more suitable for younger demographics.

Drug Efficacy Monitoring

Changes in EEG spectral patterns have been correlated with antidepressant response. The proposed framework could be adapted to track treatment efficacy longitudinally by monitoring shifts in classification confidence scores.

Chapter 8

Challenges and Limitations

- **Small Dataset Size:** The MODMA dataset contains 53 participants for the 128-channel resting condition (24 MDD, 29 controls). While adequate for demonstrating proof of concept, clinical deployment requires much larger cohorts for robust generalisation.
- **EEG Non-Stationarity:** EEG signals are highly sensitive to cognitive state, fatigue, and attention. Resting-state recordings may contain transient fluctuations that confound classification, particularly for subtypes like anxiety disorder where arousal levels are chronically elevated.
- **Electrode Discomfort:** The 128-electrode elastic cap can be uncomfortable for prolonged wear, limiting patient compliance for longitudinal studies. Future work should explore lighter, ear-based BCI setups.
- **Binary and Multi-Class Imbalance:** Within some MDD subtypes (notably OCD and anxiety disorder), the number of positive samples is substantially smaller than healthy controls, leading to inflated accuracy metrics and potentially lower minority-class recall.
- **Limited Demographic Scope:** Only age and gender are incorporated as demographic covariates. Factors such as ethnicity, socioeconomic status, comorbidities, and medication history—all known to influence EEG patterns—are not accounted for.
- **Qualitative Only Classification:** The system classifies depression into binary or subtype categories but does not estimate *severity* (mild, moderate, severe). Severity grading is clinically essential for treatment planning.
- **Simulation-Only Validation:** All evaluations are conducted offline on pre-recorded data. Real-time deployment introduces additional challenges: streaming artefact removal, latency constraints, and adaptive model updates as patient state evolves.
- **Interpretability:** Although the CNN achieves the highest accuracy, it operates as a black-box model. Clinicians require explanations of which EEG features or brain regions drive the classification decision.
- **No Cross-Dataset Validation:** The model is trained and tested exclusively on MODMA. Generalisability to other EEG datasets (different recording hardware, demographics, or clinical protocols) is not established.

Chapter 9

Conclusion

This report has presented a comprehensive analysis of the EEG-based depression detection framework proposed by Ksibi et al. (2023). The key contributions of the paper are:

1. A complete automated pipeline from raw 128-channel resting EEG to multi-class MDD subtype classification, combining preprocessing (ICA, bandpass filtering), feature engineering (8 linear + 2 nonlinear features per channel), and demographic fusion.
2. Systematic comparison of three classifiers—Random Forest, XGBoost, and 1D CNN—across six MDD subtypes on the MODMA dataset.
3. The 1D CNN achieves **97% accuracy** at 25 training epochs, outperforming prior work (CNN+GRU: 89.63%, CNN: 91.01%, ResNet-50+LSTM: 90.02%) on the same dataset.
4. Demonstration that integrating demographic covariates (age, gender) with EEG spectral and amplitude features improves classification performance, addressing a key gap in prior literature.

Limitations include small dataset size, signal non-stationarity, lack of severity estimation, and limited cross-dataset validation. Promising future directions are:

- Collecting larger, more demographically diverse EEG datasets.
- Extending to *severity* grading (mild / moderate / severe).
- Applying lightweight portable EEG hardware (ear-BCI) for continuous monitoring.
- Incorporating explainability methods (e.g. Grad-CAM, SHAP) to make CNN decisions interpretable for clinicians.
- Adding quantitative psychological assessment scores as additional predictive features.

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